RACE
Regular Analysis Through Continuous Exercise

Quadratic Equation : RACE-01

Single Correct Question

1. The roots of the equation $x^2 - 2x - 1 = 0$

(A) $\sqrt{2} \pm 1$ (B) $1 \pm \sqrt{2}$ (C) 1, 2 (D) –2, 1

MQE226

2. The real roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$ then value of p-

(A) ±5 (B) ±2 (C) ±7 (D) ±6

MQE142

3. Quadratic equation whose roots are the numbers $\frac{1}{10-\sqrt{72}}$ and $\frac{1}{10+6\sqrt{2}}$ is

(A) $2x^2 - 20x + 1 = 0$ (B) $28x^2 - 20x + 2 = 0$

(C) $28x^2 - 20x + 1 = 0$ (D) $28x^2 - 20x + 5 = 0$

MQE141

4. If α and β are the roots of $x^2 + px + q = 0$, then the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$ is–

(A) $x^2 - (p^2 - 2q)x + p^2(p^2 - 4q) = 0$ (B) $x^2 - 2(p^2 - 2q)x + p^2(p^2 - 4q) = 0$

(C) $x^2 - 2(p^2 - 2q)x + p^2(p^2 - q) = 0$ (D) $x^2 - 4(p^2 - 2q)x + p^2(p^2 - 2q) = 0$

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MQE167

5. If the sum of the squares of roots of equation $x^2 - 4x + p = 0$ is equal to 16 then the value of p is

(A) 0 (B) 1 (C) 2 (D) 3

MQE143

Multiple Correct Question

6. If the sum of the roots of the equation $x^2 + (2 - a - a^2)x - a^2 = 0$ equal to zero then the values of a is/are:

(A) –1 (B) –2 (C) 1 (D) 2

MQE144

7. If the ratio of the roots of the equation $ax^2 - (a + 3)x + 3 = 0$ equal to 1.5 then the values of a is/are:

(A) 2 (B) –2 (C) $-\frac{9}{2}$ (D) $\frac{9}{2}$

MQE145

8. If the roots of $4x^2 + 5k = (5k + 1)x$ differ by unity, then the values of k is/are

(A) 3 (B) 5 (C) $-\frac{1}{5}$ (D) $-\frac{1}{3}$

MQE146

For More Join: @IITJEE_Advanced

Integer Type Question

9. Number of values of x satisfying the equation $\frac{6-x}{x^2-4}=2+\frac{x}{x+2}$ is

MQE147

10. If α & β are the roots of equation $3x^2-5x-2=0$ and if the sum of the cubes of its roots is k, then the value of $27k$ is

MQE152

Subjective Type Question

11. Find the roots of equation $x^2-8x+16=0$

MQE148

12. Find the roots of the equation $a(x^2+1)-(a^2+1)x=0$, where $a\neq 0$.

MQE150

13. Express $x_1^3+x_2^3$ in terms of the coefficients of the equation $x^2+px+q=0$, where x_1 and x_2 are the roots of the equation.

MQE151

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	C	B	A	B,C	A,D	A,C	1	215

11. 4 **12.** $a,\frac{1}{a}$ **13.** $x_1^3+x_2^3=3pq-p^3$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Quadratic Equation : RACE-02

Single Correct Question

1. If the equation $9x^2 - 2x + a = 6 - ax$ possess equal roots then the value of a is–

(A) $20 \pm 2\sqrt{5}$ (B) $10 \pm 6\sqrt{5}$ (C) $20 \pm 3\sqrt{5}$ (D) $20 \pm 6\sqrt{5}$

MQE153

2. The values of m for which the equation $x^2 - x + m = 0$ possess no real roots is–

(A) $m \in (1, +\infty)$ (B) $m \in \left(\frac{1}{2}, +\infty\right)$ (C) $m \in \left(\frac{1}{3}, +\infty\right)$ (D) $m \in \left(\frac{1}{4}, +\infty\right)$

MQE154

3. The value of a for which one root of the equation $x^2 + (2a - 1)x + a^2 + 2 = 0$ is twice as large as the other is -

(A) –4 (B) –2 (C) –3 (D) –1

MQE155

Multiple Correct Question

4. If the equation $(k - 1)x^2 + (k + 4)x + k + 7 = 0$ has equal roots then the value of k is/are:

(A) $\frac{-22}{3}$ (B) $\frac{22}{3}$ (C) 2 (D) –2

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MQE156

5. If one of the roots of the equation $x^2 - \frac{15}{4}x + a = 0$ is the square of the other then the value of a is/are–

(A) $-\frac{125}{8}$ (B) $\frac{27}{8}$ (C) $-\frac{27}{8}$ (D) $\frac{125}{8}$

MQE157

Integer Type Question

6. The least integral value of k for which the equation $x^2 - 2(k + 2)x + 12 + k^2 = 0$ has two different real roots is

MQE158

7. Number of integer values of c for which the equation $(c - 2)x^2 + 2(c - 2)x + 2 = 0$ possess no real roots is

MQE159

Subjective Type Question

8. Nature of roots (real & distinct, real & coincident or imaginary) of equation:

$x^2 - (7 + \sqrt{3})x + 6(1 + \sqrt{3}) = 0$

MQE160

9. If $2 + \sqrt{3}$ is a root of the equation $x^2 + bx + c = 0$, where $b, c \in \mathbb{Q}$, find b, c.

MQE161

For More Join: @IITJEE_Advanced

10. Consider $f(x) = x^2 + bx + c$.

(a) Find c if $\alpha, \frac{1}{\alpha}$ are roots of $f(x) = 0$.

(b) Comment on sign of b & c, if $\alpha < 0 < \beta$ & $|\beta| > |\alpha|$, where α, β are roots of $f(x) = 0$.

MQE162

11. If α,β are the roots of the equation $ax^2 + bx + c = 0$, find the value of

(A) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ **MQE164**

(B) $\alpha^4\beta^7 + \alpha^7\beta^4$ **MQE165**

(C) $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$ **MQE166**

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ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	D	D	A	A,C	A,B	3	2

8. real & distinct. **9.** $b = -4, c = 1$

10. (a) $c = 1$; (b) b $\rightarrow$ negative, c $\rightarrow$ negative

11. (A) $\frac{b^2 - 2ac}{c^2}$ (B) $\frac{bc^4(3ac - b^2)}{a^7}$ (C) $\frac{b^2(b^2 - 4ac)}{a^2c^2}$

For More Join: @IITJEE_Advanced

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Regular Analysis Through Continuous Exercise

Quadratic Equation : RACE-03

Single Correct Question

1. If α, β are roots of $x^2 - px + q = 0$, then the quadratic equation whose root are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ and $\alpha^2\beta^3 + \alpha^3\beta^2$ is–

(A) $x^2 - p(p^4 - 5p^2q + 5p^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$
(B) $x^2 - p(p^4 - 5p^2q + q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$
(C) $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$
(D) $x^2 - p(p^4 - p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

MQE168

2. If the equation $(a + 3)(a - 1)x^2 + (a - 1)(4a + 3)x + (a^2 - 3a + 2) = 0$ has more than two roots then the value of a is–

(A) 0 (B) 1 (C) 2 (D) 3

MQE169

Integer Type Question

3. Given two quadratic equations $x^2 - x + m = 0$ and $x^2 - x + 3m = 0$, $m \neq 0$. Find the value of m for which one of the roots of the second equation is equal to double the root of the first equation.

MQE170

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Subjective Type Question

4. If $a.\frac{(x-b)(x-c)}{(a-b)(a-c)} + b.\frac{(x-c)(x-a)}{(b-c)(b-a)} + c.\frac{(x-a)(x-b)}{(c-a)(c-b)} = x$ then prove that it is an identity.

MQE171

5. Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.

MQE172

6. Find the coefficients of the equation $x^2 + px + q = 0$ such that its roots are equal to p and q.

MQE173

7. If α, β are the roots of $ax^2 + bx + c = 0$, then find the equation whose roots are

(a) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$ **MQE174**

(b) $\frac{1}{a\alpha + b}, \frac{1}{a\beta + b}$ **MQE175**

(c) $\alpha + \frac{1}{\beta}, \beta + \frac{1}{\alpha}$ **MQE176**

For More Join: @IITJEE_Advanced

8. For what value of 'a', the equation $(a^2 - a - 2)x^2 + (a^2 - 4)x + (a^2 - 3a + 2) = 0$, will have more than two solutions? Does there exist a real value of 'x' for which the above equation will be an identity in 'a' ?

MQE227

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ANSWER KEY

1.	C	**2.**	B	**3.**	–2
5.	$\frac{1}{x_1^3}+\frac{1}{x_2^3}=-\frac{27a^3+36a}{8}$			**6.**	$(p_1 = 0, q_1 = 0, p_2 = 1, q_2 = -2)$
7.	(a) $c^2x^2 + x(2ac - b^2) + a^2 = 0$				(b) $acx^2 - bx + 1 = 0$;
	(c) $acx^2 + (a + c)bx + (a + c)^2 = 0$				
8.	(a = 2; No real value of x)				

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RACE
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Quadratic Equation : RACE-04

Single Correct Question

1. Number of positive integral values of x satisfying the inequality
$\frac{(x-4)^{2017}.(x+8)^{2016}(x+1)}{x^{2016}(x-2)^3.(x+3)^5.(x-6)(x+9)^{2018}} \le 0$ is
(A) 0 (B) 1 (C) 2 (D) 3

MQE228

2. The complete solution of $\frac{x^2-1}{x+3} \ge 0$ & $x^2 - 5x + 2 \le 0$ is:
(A) $x \in \left[\frac{5-\sqrt{17}}{2}, \frac{5+\sqrt{17}}{2}\right]$ (B) $x \in \left[1, \frac{5+\sqrt{17}}{2}\right]$
(C) $x \in (-3, -1]$ (D) $x \in (-3, -1] \cup [1, \infty)$

MQE229

3. The solution of the inequality $2x - 1 \le x^2 + 3 \le x - 1$ is
(A) $x \in R$ (B) $[-2, 2]$ (C) $(-2, 2)$ (D) $x \in \phi$

MQE230

4. The number of prime numbers satisfying the inequality $\frac{x^2-1}{2x+5} < 3$ is
(A) 1 (B) 2 (C) 3 (D) 4

MQE231

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5. The number of the integral solutions of $x^2 + 9 < (x + 3)^2 < 8x + 25$ is :
(A) 1 (B) 3 (C) 4 (D) 5

MQE232

6. Number of non-negative integral values of x satisfying the inequality $\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \ge 0$ is
(A) 0 (B) 1 (C) 2 (D) 3

MQE233

Integer Type Question

7. Find the number of integers satisfying the following Inequalities

(i) $x^4 - 5x^2 + 4 \le 0$

MQE234

(ii) $x^4 - 2x^2 - 63 \le 0$

MQE235

(iii) $x^2 + 6x - 7 \le 2$

MQE236

(iv) $\frac{14x}{x+1} < \frac{9x-30}{x-4}$

MQE237

(v) $\frac{x^2+2}{x^2-1} < -2$

MQE238

For More Join: @IITJEE_Advanced

Subjective Type Question

Solve the following questions for $x \in \mathbb{R}$

8. $\frac{x^2-5x+12}{x^2-4x+5}<3$

MQE180

9. $\frac{x^2+6x-7}{x^2+1}\leq 2$

MQE181

10. $\frac{2x}{x^2-9}\leq\frac{1}{x+2}$

MQE182

11. $\frac{x^2-5x+12}{x^2-4x+5}>3$

MQE183

12. $\frac{x^2+4x+4}{2x^2-x-1}>0$

MQE184

13. Solve for $x \in \mathbb{R}$

(a) $x^2-3x+4>0$

MQE177

(b) $\frac{x}{x^2+2}<1$

MQE178

14. Solve for $x \in \mathbb{R}$

$$\frac{(2x-1)(x+3)(2-x)(1-x)^2}{x^4(x+6)(x-9)(2x^2+4x+9)}<0$$

MQE179

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ANSWER KEY

Question	1	2	3	4	5	6
Answer	D	B	D	D	D	D

7. (i) 4 (ii) 7 (iii) 9 (iv) 2 (v) 0

8. $x\in\left(-\infty,\frac{1}{2}\right)\cup(3,\infty)$

9. $x \in \mathbb{R}$

10. $x \in (-\infty,-3)\cup(-2,3)$

11. $\left(\frac{1}{2},3\right)$

12. $(-\infty,-2)\cup\left(-2,-\frac{1}{2}\right)\cup(1,\infty)$

13. (a) $x \in \mathbb{R}$; (b) $x \in \mathbb{R}$

14. $x\in(-6,-3)\cup\left(\frac{1}{2},2\right)-\{1\}\cup(9,\infty)$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Quadratic Equation : RACE-05

Single Correct Question

1. If the inequality $x^2-(k-3)x-k+6>0$ is valid for all real x then the values of k is–
(A) $(3,-5)$ (B) $(-3,5)$ (C) $(3,5)$ (D) $(-3,-5)$
MQE194

2. Find the range of $\frac{x^2+2x+1}{x^2+2x+7}$, where x is real
(A) $[0,1)$ (B) $[0,2)$ (C) $[1,3)$ (D) $[1,2)$
MQE195

3. If x is real, then the range of $\frac{x^2-x+1}{x^2+x+1}$ is–
(A) $[1,3]$ (B) $[1,2]$ (C) $\left[\frac{1}{3},3\right]$ (D) $\left[\frac{1}{2},2\right]$
MQE196

4. If x be real then the range of $\frac{x}{x^2-5x+9}$ is–
(A) $\left[-\frac{1}{10},1\right]$ (B) $[-1,1]$ (C) $[-1,3]$ (D) $\left[-\frac{1}{11},1\right]$
MQE197

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Multiple Correct Question

5. If $x^2+bx+c=0$ & $2x^2+9x+10=0$ have both roots in common then the correct statement is/are–
(A) $b=\frac{9}{2}$ (B) $c=3$ (C) $b=\frac{7}{2}$ (D) $c=5$
MQE198

6. If $x^2-7x+10=0$ & $x^2-5x+c=0$ have a common root then the values of c is/are–
(A) 1 (B) 0 (C) 6 (D) 2
MQE199

Integer Type Question

7. The absolute value of a for which the equations $x^2+ax+1=0$ and $x^2+x+a=0$ have exactly one root in common is
MQE200

8. Find the sum of the minimum value of $y_1=x^2+2x+2$ and $y_2=4x^2-16x+15$
MQE201

9. For what integral k is the inequality $x^2-2(4k-1)x+15k^2-2k-7>0$ valid for any real x ?
MQE202

10. For what least integral k is the expression $(k-2)x^2+8x+k+4>0$ for all values of x ?
MQE203

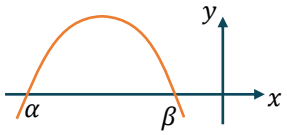
11. If the greatest value of $\frac{x+2}{2x^2+3x+6}$ for real values of x is $\frac{k}{3}$, then the value of k is
MQE204

For More Join: @IITJEE_Advanced

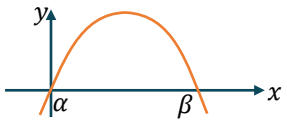
Subjective Type Question

12. For following graphs of $y = ax^2 + bx + c$ with $a, b, c \in R$, comment on the sign of :

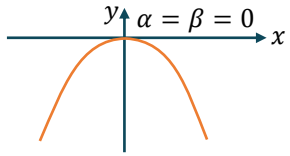
(i) a (ii) b (iii) c (iv) D (v) $\alpha + \beta$ (vi) $\alpha\beta$

(1) 

MQE205

(2) 

MQE206

(3) 

MQE207

13. Given the roots of equation $ax^2 + bx + c = 0$ are real & distinct, where $a, b, c \in R^+$, then the vertex of the graph will lie in which quadrant.

MQE208

14. Find the range of 'a' for which :

(a) $ax^2 + 3x + 4 > 0 \quad \forall\, x \in \mathbb{R}$

MQE209

(b) $ax^2 + 4x - 2 < 0 \quad \forall\, x \in \mathbb{R}$

MQE210

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15. The trinomial $ax^2 + bx + c$ has no real roots, $a + b + c < 0$. Find the sign of the number c.

MQE211

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	B	A	C	D	A,D	B,C	2	0	3	5	1

12. (1) (i) $a < 0$ (ii) $b < 0$ (iii) $c < 0$ (iv) $D > 0$ (v) $\alpha + \beta < 0$ (vi) $\alpha\beta > 0$

(2) (i) $a < 0$ (ii) $b > 0$ (iii) $c = 0$ (iv) $D > 0$ (v) $\alpha + \beta > 0$ (vi) $\alpha\beta = 0$

(3) (i) $a < 0$ (ii) $b = 0$ (iii) $c = 0$ (iv) $D = 0$ (v) $\alpha + \beta = 0$ (vi) $\alpha\beta = 0$

13. Third quadrant **14.** (a) $a > 9/16$ (b) $a < -2$ **15.** $c < 0$

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RACE
Regular Analysis Through Continuous Exercise

Quadratic Equation : RACE-06

Single Correct Question

1. If α, β are roots of $7x^2 + 9x - 2 = 0$ where $\alpha < \beta$ then select the incorrect statement –
(A) α and β are greater than –3
(B) '0' lies in between the roots α and β.
(C) α and β are less than 1.
(D) α and β are greater than 1.

MQE212

2. If $a > 1$, roots of the equation $(1-a)x^2 + 3ax - 1 = 0$ are -
(A) one positive one negative (B) both negative
(C) both positive (D) both non-real

MQE213

3. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3 then find the values of a is–
(A) $a < 2$ (B) $a > 2$ (C) $a < 3$ (D) $a > 3$

MQE214

4. If α, β are the roots of $x^2 - 3x + a = 0, a \in \mathbb{R}$ and $\alpha < 1 < \beta$, then find the values of a is–
(A) $a < 2$ (B) $a > 2$ (C) $a < 3$ (D) $a > 3$

MQE215

5. If two roots of $x^3 + 3x^2 - 9x + c = 0$ are equal, then the values of c is–
(A) $-28, 3$ (B) $-21, 2$ (C) $-27, 5$ (D) $-17, 2$

MQE216

Integer Type Question

6. If α, β are roots of $4x^2 - 16x + \lambda = 0, \lambda \in \mathbb{R}$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then number of integers point in the range of λ.

MQE217

7. If roots of the equation $(2-x)(x+1) = a$ are real and positive for $a \in \left(k, \frac{l}{4}\right]$ then the value of $l - k$ is

MQE218

Subjective Type Question

8. If x and y are two real quantities connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then show that x will lie between 3 and 6. and y between 1 and 10.

MQE219

9. Let α, β be two of the roots of the equation $x^3 - px^2 + qx - r = 0$. If $\alpha + \beta = 0$, then show that $pq = r$

MQE220

For More Join: @IITJEE_Advanced

10. If α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$, then find the value of

(a) $\sum\alpha^2$

MQE221

(b) $\sum\frac{1}{\alpha}$

MQE222

(c) $\sum\alpha^2(\beta+\gamma)$

MQE223

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ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	D	C	A	A	C	3	7

10. (a) $\frac{1}{a^2}(b^2-2ac)$ (b) $-\frac{c}{d}$ (c) $\frac{1}{a^2}(3ad-bc)$

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